

General Sir John Kotelawala Defence University



Faculty of Computing Essentials of Artificial Intelligence

AI-Based Legal Case Law Recommendation System Using Case Facts for Sri Lankan Criminal Law

Group Number - 14	
Index	Name
D/DBA/25/0009	MT Jayaweera
D/BIS/24/0040	K.S.Navodya
D/BIT/24/0052	WST Gunawardana
D/COE/25/0014	Pabadi Dayawansha

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Background of the Selected Problem

In the field case law is about looking at what courts have decided before when dealing with similar issues or situations. When lawyers get a case they look at what happened in other cases to see how the courts handled things. These old decisions help lawyers do their research and get ready for arguments. Law students also study case law to understand how laws work in life and how they are interpreted.

When doing research you have to look at a lot of old cases to find the ones that are relevant to what you are working on. Lawyers and law students used to use books and databases to find these cases. They had to find the cases and then look at what happened and what the court decided to see if it applied to their case.

You cannot just look for cases with the words to find relevant ones. Cases that are similar might use words or describe things differently. So to do research you have to understand what is similar about the cases not just look for the same words.

Now computers and Artificial Intelligence can help find information and make it easier to search through a lot of documents.. When using Artificial Intelligence for legal research you have to make sure the information is real and trustworthy because lawyers and law students need to use real court decisions.

For this project we are only looking at case law in Sri Lanka that deals with cases. The people who will use this system are lawyers and law students. It is meant to help them with their work and their studies. We want case law to be helpful for both lawyers who are working on cases and law students who are learning about the law. The system is for case law, in Sri Lanka. It should help lawyers and law students with their research.

Problem Statement

The big issue that this project is trying to solve is that it is really hard to find Sri Lankan criminal case laws that're relevant to a particular case.

Lawyers and law students have to look through a lot of cases to find ones that are similar to the one they are working on. This takes a lot of time. They have to be good at picking the right words to search for and spotting the right cases.

The normal way of searching using keywords does not always work because it might not find cases that're similar if they use different words to describe the same thing.

Some tools that use intelligence can help with legal research faster but they can also give wrong information that is not based on real legal sources like made up case names or facts.

So we need a system that can find and suggest cases that are similar to the one being researched and get the information from a trusted collection of Sri Lankan criminal case laws.

This system should be able to do this by looking at how similar the facts of the cases are and it should only use real Sri Lankan criminal case laws.

Objectives of the Proposed AI Solution

The main goal of the proposed Artificial Intelligence solution is to create an Artificial Intelligence Based Legal Case Law Recommendation System Using Case Facts that helps lawyers and law students find Sri Lankan criminal case laws quickly.

The specific goals of the Artificial Intelligence solution are:

- To create a list of Sri Lankan criminal case laws with information like case ID, case name, case facts and judgments.
- To get the case law data ready for training and testing the proposed Artificial Intelligence model.
- To. Train a neural network model that can find similarities between the facts of a case that a user gives and the facts of old Sri Lankan criminal cases.
- To make it easy for lawyers and law students to enter the facts of a case or a case they are researching.
- To suggest old Sri Lankan criminal cases by comparing the facts of the users case with the cases in the list.
- To give information about the suggested cases, including the case ID, case name, important case facts and judgment so users can look at the cases more closely.
- To help lawyers do their research for work by saving them time and effort when looking for old criminal cases that might be relevant.
- To help law students with their studies and research by finding criminal cases that are relevant to what they are learning.
- To reduce the chance of legal information by only suggesting cases that are, in the systems list and not making up case names, facts or judgments.
- To see how well the proposed Artificial Intelligence solution works at finding and suggesting Sri Lankan criminal case laws based on case facts.

Target Users

The primary targeted end-users of the proposed AI Based Legal Case Law Recommendation system are lawyers and law students who need to be able to access relevant Sri Lankan criminal case laws efficiently.

Lawyers:

The system will provide a tool for lawyers to carry out legal research by inputting the facts of a current criminal case and retrieve past Sri Lankan criminal cases that have facts in common. This can save time and effort in reviewing a big volume of case law in a manual manner.

Law Students:

The system will help law students in academic research and learning by enabling them to search for previous criminal cases from a fact pattern. A set of recommended cases are provided for students to further explore their facts and judgments.

The proposed system is designed as a legal research tool, not as a legal aid provider, as it will not give legal advice, legal applicability of a case or predict court decisions.

Proposed AI Techniques

The AI-Based Legal Case Law Recommendation System will be based on Natural Language Processing (NLP), Siamese Neural Network, Text Embedding and Similarity Measurement techniques to identify and recommend Sri Lankan Criminal Case Laws based on factual similarity between cases.

1. Natural Language Processing (NLP)

The textual case-fact data will be preprocessed using the NLP techniques before feeding inputs to the neural network. The preprocessing step will involve cleaning and normalizing the facts from the cases and format the text for model processing.

2. Siamese Neural Network

The learning approach for the similarity between the pairs of case facts will be based on Siamese Neural Network (SNN), as it is the major artificial network technique used.

The network will receive two case-fact inputs and have a shared set of neural network layers that will produce two numerical representations of the case (embeddings). The model will be trained to generate similar representations for cases having similar factual situations and different representations for cases having dissimilar situations.

The trained network can then be used to create an embedding for a new inputted case and compare it with the embeddings of the historical cases in the dataset.

3. Text Embeddings

The neural network will convert the case facts into a numerical vector representation of the facts. These embeddings will look for key features of the text and enable cases to be mathematically compared.

The embedding created for the criminal case presented to the user will be compared to the embedding of historical Sri Lankan criminal cases.

4. Cosine Similarity

The similarity between historical cases and the user's case will be computed as Cosine similarity.

The similarity scores will be used to rank the historical cases from the most similar to the least similar.

For example:

Case A $\rightarrow$ 0.91

Case B $\rightarrow$ 0.86

Case C $\rightarrow$ 0.79

Case D $\rightarrow$ 0.64

The system can then suggest the Top-K cases that are most similar.

Similarity score will reflect similarities of case facts, and not be understood as legal applicability or legal correctness.

5. TFIDF and Cosine Similarity as a baseline .

A baseline model using traditional TF-IDF + Cosine Similarity approach will be implemented.

The baseline will be used to compare the performance of:

Traditional keyword-based similarity

Against: Neural-network-based semantic similarity

An evaluation of the retrieval of factually similar criminal cases will be made by comparing this approach with the proposed approach using a neural network.

Justification for Selecting the AI Techniques

1. Justification for NLP

The legal case law is of unstructured text based content. Hence, NLP methods have to be used to clean and prepare the facts of the case before they can be fed into the AI model.

NLP is used to preprocess raw legal text and make it suitable for use in the neural network.

2. Justification for Siamese Neural Network

A Siamese Neural Network is chosen as it is designed to measure similarity between two cases instead of classify a case into a pre-defined category is the primary aim of the proposed system.

In traditional classification neural networks, the system would have to guess the classes of events that are previously defined, like "murder," "theft," or "assault. But the proposed system has to decide if the factual situation of two cases is comparable.

Siamese Neural Network is a particular type of neural network that is suitable for similarity and comparison tasks. It can get case facts represented and determine relationships between similar and dissimilar case pairs.

Thus, it is applicable to the suggested legal case-law recommendation problem.

3. Justification for Text Embeddings

Text embeddings are numerical vectors where all the textual data within the case facts is embedded. This makes it possible for the system to mathematically compare different cases.

Embeddings are especially helpful for this project, as similar cases could have different words or word orders. More meaningful relationships can be captured between cases than just exact matching of keywords by a suitable neural-network based representation.

4. Justification for Cosine Similarity

The cosine similarity is chosen as an effective way to compare the vector representations generated by the neural network.

After obtaining the case facts and converting them to embeddings, the cosine similarity can identify the similarity between the representations of two cases.

The scores obtained can then be used to rank historical cases to recommend cases.

5. Justification for TF-IDF Baseline

TF-IDF is used as a baseline, as it is a traditional information-retrieval approach that would be used if the neural-network approach is not employed.

This is an important point because the project should not just show that it is possible to get a neural network to retrieve cases, but also show that it is possible to get it to retrieve them reliably. It should also examine the possibility that the neural-network approach is an improvement over a simpler traditional approach.

Expected System Inputs and Outputs

Inputs

- User entered case facts - case entered by a lawyer or a law student through the system interface Textual description of facts in the current .
- Data set of historical case law - The already per-collected data set of criminal cases from Sri Lanka consisting of case ID, case name, case facts, and judgment which was referred to by your team members in the "Objectives" part. This data set is loaded before querying the system.

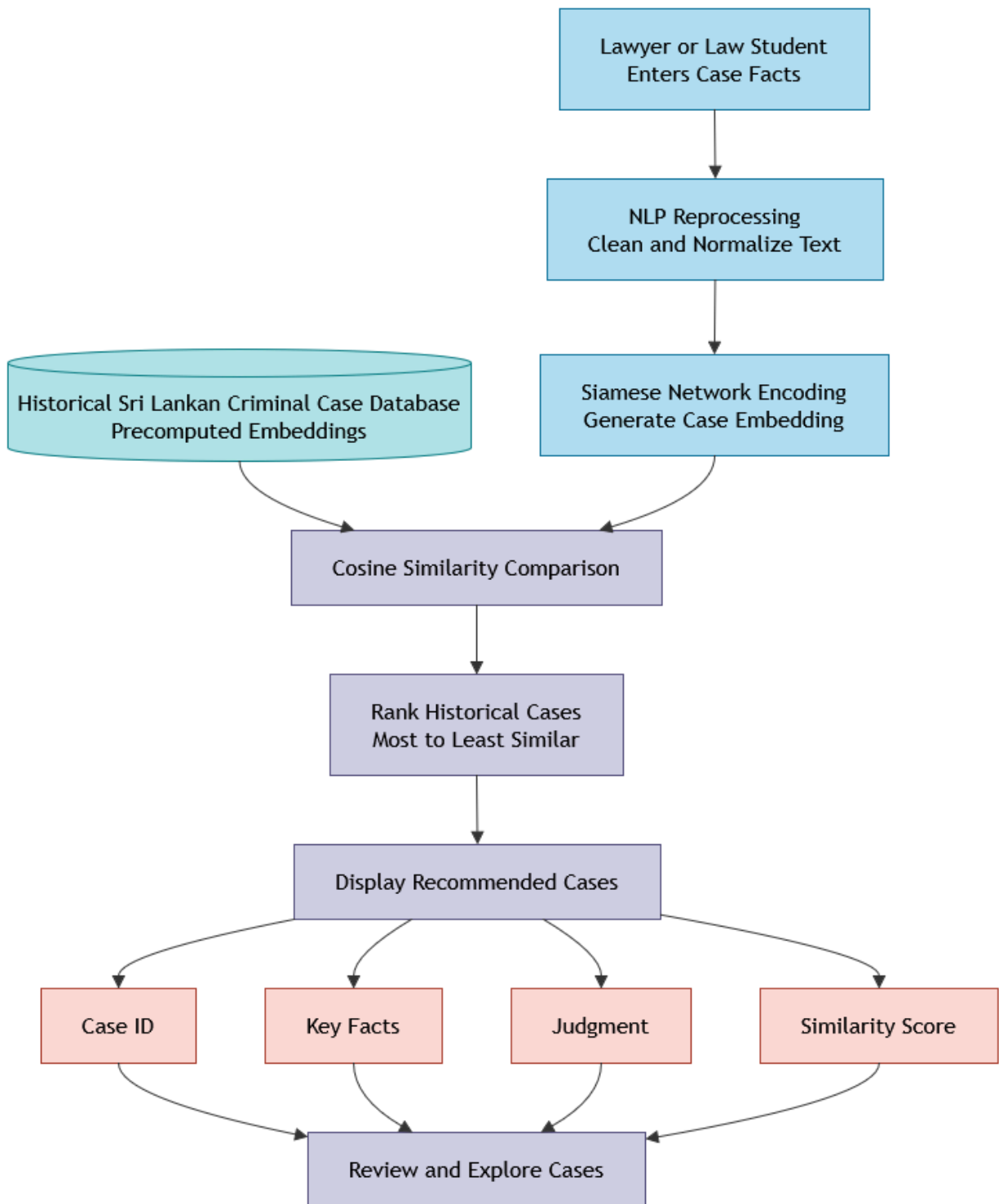
Outputs

- A ranked list of Top-K similar cases, ordered by cosine similarity score (highest to lowest), matching the example your teammates gave (Case A $\rightarrow$ 0.91, Case B $\rightarrow$ 0.86, etc.).
- For each recommended case: case ID, case name, key case facts, and judgment, so the user can review it without leaving the system.
- The similarity score displayed alongside each result, clearly labeled as a factual-similarity measure — not a legal-relevance or predicted-outcome score (to stay consistent with the disclaimer already in the proposal).

Proposed Tools and Technologies

Category	Tool/Technology	Purpose
Programming language	Python	Core language for NLP + ML pipeline
NLP prepossessing	spaCy / NLTK	Cleaning, tokenizing, normalizing case-fact text
Deep learning framework	TensorFlow /Keras	Building and training the Siamese Neural Network
Baseline model	scikit-learn (TfidfVectorizer + cosine_similarity)	TF-IDF baseline comparison
Similarity computation	scikit-learn / NumPy	Cosine similarity between embedding
Dataset storage	MongoDB / MySQL / PostgreSQL	Storing case ID, name, facts, judgment
Data collection	Manual collection / BeautifulSoup (if scraping public judgments)	Building the Sri Lankan criminal case dataset
Interface	Flask or Django (backend) + HTML/React (frontend)	Letting users enter case facts and view results
Development tools	Jupyter Notebook, VS Code, Git/GitHub	Development and version control
Environment	Google Colab	GPU access for training the Siamese Network

Initial System Workflow



Work Division Among Members

The works to design and develop the AI-Based Legal Case Law Recommendation System are divided among the four group members according to the major components of the system. This means that each member has a clear, defined responsibility but at the same time, each member contributes collectively to the proposal, the progress review, the final report and the presentation as required by the module guidelines.

Member	Degree Programme	Primary Responsibility
Pabadi Dayawansa (D/COE/25/0014)	COE	Data Collection and Dataset Preparation - collecting Sri Lankan criminal case judgements, extracting case ID, case name, case facts and judgement for each entry and preparing the dataset to train and test the AI model.
MT Jayaweera (D/DBA/25/0009)	DBA	NLP Preprocessing and Baseline Model - Cleaning and normalising the case-fact text, and implementing the TF-IDF and Cosine Similarity baseline model used for benchmarking the neural network approach.
K.S. Navodya (D/BIS/24/0040)	BIS	Siamese Neural Network development - From design to building and training the Siamese Neural Network, preparing the case-fact pairs for training and generating the embeddings used for similarity comparison.
WST Gunawardana (D/BIT/24/0052)	BIT	System interface and integration - Building the backend and frontend interface for inputting case facts and viewing recommended cases, and integrating the trained model with the case law database.

Shared responsibilities: All four members will work together to test the system, compare its performance with the baseline, discuss ethical and social issues and write the proposal, progress review, final report and presentation. This allocation of work will be reviewed and updated at the Progress Review stage (Week 8) and reflected in the individual contribution sheet submitted with the Final Submission (Week 10).

Timeline for Completing the Work

The project will be completed during the 10-week period allocated to the module and structured around the three assessment stages: Proposal (Week 4), Progress Review (Week 8) and Final Submission (Week 10).

WEEK	PHASE	KEY ACTIVITIES
1	Problem Identification & Planning	Confirm topic, lecturer approval, main responsibilities, shared documentation, finalise topic
2	Background Research	Review of existing legal research methods in Sri Lanka, review of related AI based legal recommendation systems, define the problem statement and objectives
3	Proposal Drafting	Background, objectives, target users, justification of AI technique, tools and technologies, and initial workflow diagram
4	Proposal Submission	Discuss proposal within group members, finalise & submit proposal
5	Data Collection	Start reviewing criminal case judgements from Sri Lanka; extract case ID, case name, facts and judgement; construct initial dataset structure.
6	Data Preparation & NLP Preprocessing	Text cleaning pipeline; case-fact text clean and normalise, case-fact pairs training
7	Baseline & Model Development	Implement TF-IDF + Cosine Similarity baseline, start building and training Siamese Neural Network

8	Progress Review	Submit completed AI techniques, dataset details, network structure, and partial implementation screenshots. Problems encountered should be documented
9	System Integration & Testing	Integrate the trained model with the backend/frontend interface. Test with sample case facts. Compare the neural network results with the TF-IDF baseline.
10	Final Submission & Presentation	Final report, ethical/social considerations and limitations, presentation slides and video, individual contribution sheet

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